

Multiple Random Variables Exercises

Mathematics for AI

Question 1. Marginals of a joint Bernoulli pmf

Let X and Y both be random variables with the Bernoulli distribution $X \sim B(p_1)$, $Y \sim B(p_2)$, with the following joint pmf:

$$\begin{aligned} f_{X,Y}(0,0) &= \frac{1}{10} & f_{X,Y}(0,1) &= \frac{3}{10} \\ f_{X,Y}(1,0) &= \frac{3}{20} & f_{X,Y}(1,1) &= \frac{9}{20} \end{aligned}$$

(a)

Calculate p_1 .

All the given values are not negative and sum to 1, so they form a valid probability mass function (pmf):

$$\frac{1}{10} + \frac{3}{10} + \frac{3}{20} + \frac{9}{20} = 1$$

Since $X \sim B(p_1)$, then the parameter is $p_1 = P(X = 1) = f_X(1)$. We need to calculate the marginal pmf of X . To do this we need to marginalise the joint pmf over the values of Y :

$$f_X(x) = \sum_y f_{X,Y}(x, y)$$

Since Y is Bernoulli it can take values of $\{0, 1\}$, so we can work out p_1 as follows:

$$\begin{aligned} p_1 = f_X(1) &= f_{X,Y}(1, 0) + f_{X,Y}(1, 1) \\ &= \frac{3}{20} + \frac{9}{20} \\ &= \frac{12}{20} = \frac{3}{5} \end{aligned}$$

Quick sanity check X can only take values of 0 and 1, so $f_X(0) + f_X(1) = 1$.

$$\begin{aligned} f_X(0) &= f_{X,Y}(0, 0) + f_{X,Y}(0, 1) \\ &= \frac{1}{10} + \frac{3}{10} \\ &= \frac{4}{10} = \frac{2}{5} \end{aligned}$$

and $\frac{3}{5} + \frac{2}{5} = 1$ so our p_1 answer makes sense.

(b)

Calculate p_2 .

Following a similar approach to p_1 but now we need to calculate the marginal pmf of Y . To do this we need to marginalise the joint pmf over the values of X . Since $Y \sim B(p_2)$, then the parameter is $p_2 = P(Y = 1) = f_Y(1)$.

$$f_Y(y) = \sum_x f_{X,Y}(x, y)$$

Since X is Bernoulli it can take values of $\{0, 1\}$, so we can work out p_2 as follows:

$$\begin{aligned} p_2 = f_Y(1) &= f_{X,Y}(0, 1) + f_{X,Y}(1, 1) \\ &= \frac{3}{10} + \frac{9}{20} \\ &= \frac{15}{20} = \frac{3}{4} \end{aligned}$$

Quick sanity check Y can only take values of 0 and 1, so $f_Y(0) + f_Y(1) = 1$.

$$\begin{aligned} f_Y(0) &= f_{X,Y}(0, 0) + f_{X,Y}(1, 0) \\ &= \frac{1}{10} + \frac{3}{20} \\ &= \frac{5}{20} = \frac{1}{4} \end{aligned}$$

and $\frac{1}{4} + \frac{3}{4} = 1$ so our p_2 answer makes sense.

Question 2. Geometric reasoning with a uniform joint density

Let X, Y both be random variables uniformly distributed on $[-1, 1]$. We can think of their joint density function $f_{X,Y}$ as describing the likelihood of a point selected randomly in the square with corners $(1, 1), (1, -1), (-1, -1), (-1, 1)$. Since the area of this square is 4, the density function is $f_{X,Y}(x, y) = 0.25$, i.e. the chance of the point landing in a space with unit area is 0.25. Using this insight, calculate the following probabilities:

My understanding:

Since X and Y are independent, the pair (X, Y) can take any value in $[-1, 1] \times [-1, 1]$, i.e. the **square** with corners $(\pm 1, \pm 1)$ and area 4. The density is constant on this square, so:

$$f_{X,Y}(x, y) = \frac{1}{4} \quad \text{for } (x, y) \in [-1, 1]^2,$$

and 0 everywhere else. For any region A we thus have:

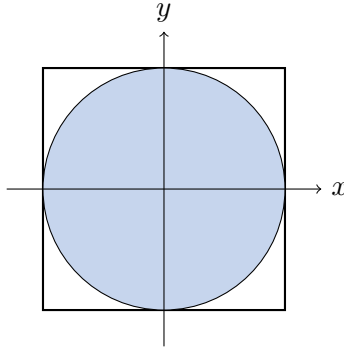
$$P((X, Y) \in A) = \iint_A f_{X,Y}(x, y) dx dy = \frac{1}{4} \text{area}(A \cap [-1, 1]^2),$$

Note in the above Area is a function that takes input region A which with the intersection, we only consider its parts within $[-1 \text{ and } 1]^2$, the Cartesian-product of $[-1 \text{ and } 1] \times [-1 \text{ and } 1]$ so each of these probabilities just needs us to find an area.

(a)

$$P(X^2 + Y^2 \leq 1)$$

So for $P(X^2 + Y^2 \leq 1)$:



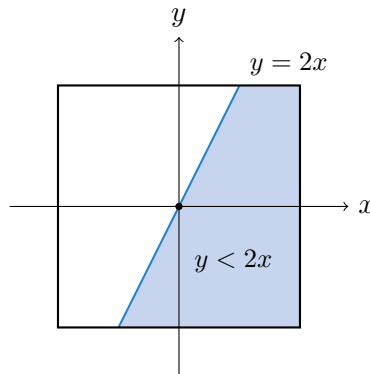
$X^2 + Y^2 \leq 1$ represent a disc of radius 1 centred at the origin, since $\sqrt{x^2 + y^2}$ is the distance from the origin to (x, y) by the Pythagoras Theorem. The disc touches the sides of the square at $(\pm 1, 0)$ and $(0, \pm 1)$ but does not extend beyond it, so $A \cap [-1, 1]^2 = A$ and no trimming is needed.

So the area we are looking for is the area of the disc which is πr^2 and from the values of X and Y , $r = 1$ so the area is simply π .

$$\text{So } P(X^2 + Y^2 \leq 1) = \frac{\pi}{4}$$

(b)

$$P(2X - Y > 0)$$



So the area we are looking for is the area that sits under the line $y = 2x$. We need it to be the area underneath as $y < 2x$ and if we test a point e.g. $(1, 0)$ then $0 - 2(1)$ is less than 0, which confirms we are considering the correct side of the line.

Now for the area, the line $y = 2x$ passes through the origin which also is the centre of our square and any line through the centre of a square divides its area into two equal halves. We know the square has an area of 4 so our area is $\frac{4}{2} = 2$

$$\text{So } P(2X - Y > 0) = \frac{1}{4} \cdot 2 = \frac{1}{2}$$

(c)

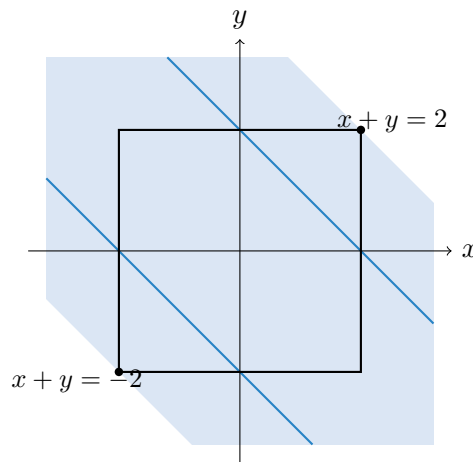
$$P(|X + Y| < 2)$$

Since $-1 \leq X \leq 1$ and $-1 \leq Y \leq 1$, adding gives $-2 \leq X + Y \leq 2$, so $|X + Y| \leq 2$ everywhere in the square. The inequality is strict except where $|X + Y| = 2$, which requires $x + y = \pm 2$; since x and y are each at most 1 in magnitude, this forces $x = y = 1$ or $x = y = -1$. So the condition $|X + Y| < 2$ fails only at the two corners $(1, 1)$ and $(-1, -1)$. Two points have zero area, so

$$\text{area}(A \cap [-1, 1]^2) = 4 - 0 = 4.$$

$$\text{So } P(|X + Y| < 2) = \frac{1}{4} \cdot 4 = 1.$$

We can also validate this through graphing the two lines that clip the square at $(1, 1)$ for $y = -x + 2$ and $(-1, -1)$ for $y = -x - 2$:



Quick note just for me on area vs density:

These are three distinct functions with three distinct kinds of input.

The density $f_{X,Y}$ is evaluated at a point and measures how thickly probability is packed there.
area is evaluated on a region and measures how much space that region occupies.

They come together inside the integral, where the region determines what to sweep over and the density determines what to accumulate.

When the density is constant it factors out, leaving $P = \text{density} \times \text{area}$.

Object	Takes	Returns
$f_{X,Y}$	a point (x, y)	a density, here $\frac{1}{4}$
$\text{area}(\cdot)$	a region	a number, e.g. π
$P(\cdot)$	an event	a probability in $[0, 1]$

Question 3. Departure time and journey length

I leave for work between 8AM and 8:30AM and it takes between 40 and 50 minutes to get there. Assume that departure time and journey length are independent and each is uniformly distributed. What is the probability that I will arrive before 9AM?

Set-up.

Let D be the minutes after 8AM so $D \sim U[0, 30]$ Let T be the time for my journey to work in minutes so $T \sim U[40, 50]$ If $D + T < 60$ I would get to work by 9AM

So we are looking for the $P(D + T < 60)$

Working.

Since D and T are uniformly distributed, their densities are constant on their respective supports, so $f_D(x) = \frac{1}{30}$ on $[0, 30]$ and $f_T(y) = \frac{1}{10}$ on $[40, 50]$.

Since D and T are independent, the joint density is the product of the individual densities,

$$f_{D,T}(x, y) = f_D(x) f_T(y) = \frac{1}{30} \cdot \frac{1}{10} = \frac{1}{300}, \quad (x, y) \in [0, 30] \times [40, 50],$$

and zero otherwise.

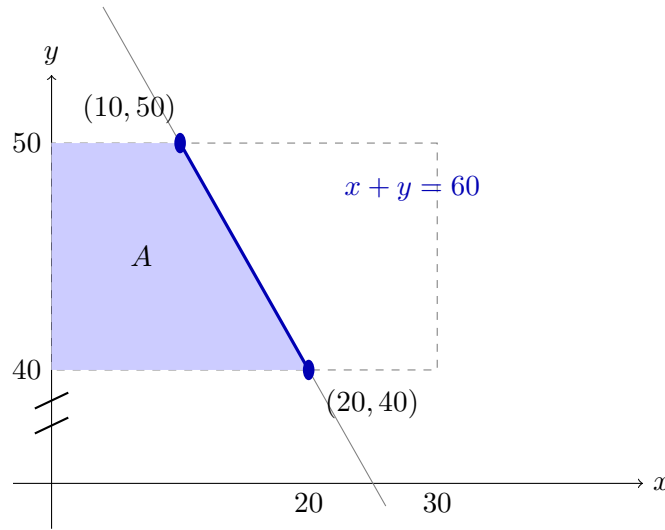


Figure 1: The support $[0, 30] \times [40, 50]$ (dashed) and the region A where $D + T < 60$ (shaded). The boundary meets the top and bottom edges, so every horizontal slice is clipped in the same way.

The event $D + T < 60$ restricts x to $[0, 60 - y]$. For $y \in [40, 50]$ we have $60 - y \in [10, 20]$, which lies inside $[0, 30]$, so the upper limit is $60 - y$ throughout and the region requires no splitting. Hence

$$\begin{aligned} P(D + T < 60) &= \int_{40}^{50} \int_0^{60-y} \frac{1}{300} dx dy \\ &= \frac{1}{300} \int_{40}^{50} (60 - y) dy \\ &= \frac{1}{300} \left[60y - \frac{y^2}{2} \right]_{40}^{50} \\ &= \frac{1}{300} (1750 - 1600) = \frac{150}{300} = \frac{1}{2}. \end{aligned}$$

So I arrive before 9AM with probability $\frac{1}{2}$.

As a check, the density is constant, so the probability is the area of the region divided by 300. The region is a trapezium whose area is $A = \frac{1}{2}(a + b)h$ where a and b are the lengths of the two parallel sides. Our trapezium has parallel sides of length 20 and 10 and height 10, giving area $\frac{1}{2}(20 + 10)(10) = 150$, and $150/300 = \frac{1}{2}$ as above.

Declaration:

I used Claude Opus 5 to help me learn how to chart the graphs in LaTeX for Overleaf.